OPTICS · SEMESTER 5 · MODULE 3

Prepare: Diffraction

Everything you need before you open the question paper, from Plus Two up to Ghatak §18.1–18.2, 18.6–18.8, 20.1–20.3, 20.6 (plus circular apertures and resolution, which the past papers ask).

File 1 of 3	**Prepare** (this file): learn the ideas
File 2	Model paper: PYQs + new questions
File 3	Answer key: full solutions

Each answer in File 3 starts with topic codes such as T6 P2 M1. If a code is new to you, read that topic here first.

How to read this file

One topic at a time, in order.

REMIND ME

Facts from Class 10, Plus One or Plus Two that you have probably forgotten.

MATHS TOOL

A small piece of maths, explained the first time it is needed.

THE IDEA

The physics in plain words, before any symbols.

RESULT

What you must remember and write in the exam.

EXAM TRAP

A common mistake, and how to avoid it.

CHECK YOURSELF

Questions with the answer just below. Cover it with your thumb first.

One rule for the whole module

ADD ARROWS, THEN SQUARE

Every wave is an arrow (length = amplitude, angle = phase). To find the light at a point, add all the arrows arriving there head to tail. The brightness is the **length of the total arrow, squared**. Every formula in this module is this rule applied carefully.

Codes and map

P1–P4 Plus One / Plus Two bridge

M1–M4 Maths tools

T1–T13 Module 3 topics

Q12 question 12 of the model paper

SYLLABUS UNIT	TOPICS
12. Single-slit Fraunhofer pattern	T1, T2, T3 (+ T4 circular hole)
13. Two-slit Fraunhofer pattern	T5
14. N-slit pattern and grating	T6, T7, T8, T9
15. Fresnel diffraction, zone plate	T10, T11, T12
16. Diffraction by a straight edge	T13

15 of the 70 external marks. The single slit was a 10-mark essay in four different years; grating resolving power and half-period zones are asked almost every year.

The whole module in one story

THE IDEA

1. Light bends round edges and spreads out of openings, because it is a wave (T1).

2. A slit = many tiny sources. Adding their arrows gives the single-slit pattern; a round hole gives rings (T2–T4).

3. Two slits, then N slits: sharp bright lines. Many slits = a grating, which separates colours and can tell close wavelengths apart (T5–T9).

4. With a nearby source and screen (Fresnel), cut the wavefront into rings (zones). This explains why light seems to travel straight, a bright spot behind a disc, and the zone plate "lens" (T10–T12).

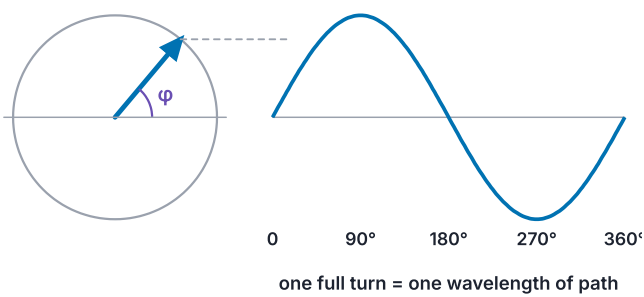
5. Behind a straight edge, the shadow is not sharp: the Cornu spiral gives the fringes (T13).

P1 Phase, path and intensity

The three wave ideas used on every page.

REMIND ME · PLUS ONE CH 14

- Wave: $y = a\cos(kx - \omega t)$, $k = 2\pi/\lambda$. Light: λ ≈ 5000 Å $= 5 \times 10^{-5}$ cm.
- Path difference Δ ⟹ phase difference $\delta = \frac{2\pi}{\lambda}\Delta$. Δ = mλ: in step (bright). Δ = (m + ½)λ: opposite (dark).
- Intensity ∝ amplitude²: $I \propto a^2$.



one full turn = one wavelength of path

Phase is an angle. A point going round the circle gives the wave (its height). One full turn (360° = 2π) is one wavelength. A path difference of λ/4 is a phase difference of 90°.

CHECK YOURSELF

Two equal waves arrive with a path difference of λ/2. Resulting intensity?

Phase difference π: the arrows point opposite ways and cancel. Intensity zero.

P2 Huygens' principle and wavefronts

> Every point of a wave acts like a new little source.

REMIND ME · PLUS TWO CH 10

- A **wavefront** is a surface on which all points vibrate in step (same phase). Far from a point source it is nearly flat: a **plane wave**.
- **Huygens**: every point on a wavefront sends out small secondary wavelets; the new wavefront is the surface touching them all.
- A lens brings a parallel beam to a point in its focal plane; a beam at angle θ to the axis focuses at height $f \tan\theta \approx f\theta$.

THE IDEA

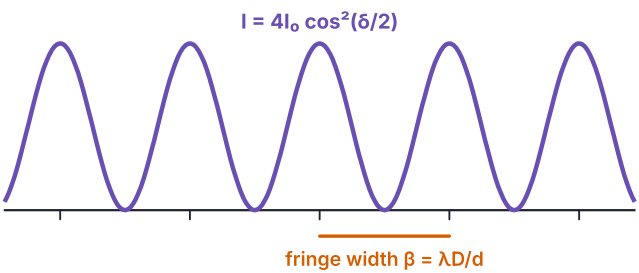
Fresnel added one thing to Huygens: the wavelets **interfere** with each other. So to find the light at any point P, add up the wavelets from every point of the wavefront, with their phases. That is diffraction theory in one sentence.

P3 Young's double slit (as in Plus Two)

> Needed for the two-slit and N-slit patterns.

REMIND ME · PLUS TWO CH 10

- Two slits d apart. At angle θ, path difference $d\sin\theta$.
- Bright: $d\sin\theta = m\lambda$. Dark: $d\sin\theta = (m+\frac{1}{2})\lambda$.
- Fringe width on a screen at distance D: $\beta = \lambda D/d$. Intensity $I = 4I_0\cos^2(\delta/2)$.



Intensity on the screen. Bright fringes ($4I_0$) where $\delta = 0, 2\pi, \ldots$; dark (zero) where $\delta = \pi, 3\pi, \ldots$. Neighbouring bright fringes are β apart, all equally spaced.

P4 Diffraction in Plus Two

> What NCERT already told you. This module proves it and goes further.

REMIND ME · PLUS TWO §10.6

- A single slit of width a gives a broad central bright band with dark bands at $a \sin\theta = n\lambda$ ($n = \pm 1, \pm 2, \ldots$), and weaker bands beyond.
- The narrower the slit, the wider the pattern.
- Resolving power: two objects are just resolved when the centre of one diffraction pattern falls on the first dark ring of the other. For a circular lens of diameter D: smallest angle $\approx 1.22\lambda/D$.

EXAM TRAP

In this module Ghatak calls the slit width b (not a) and the slit spacing d. For a single slit, $b\sin\theta = m\lambda$ gives **dark** places; for slits, $d\sin\theta = m\lambda$ gives **bright** places. Same look, opposite meaning.

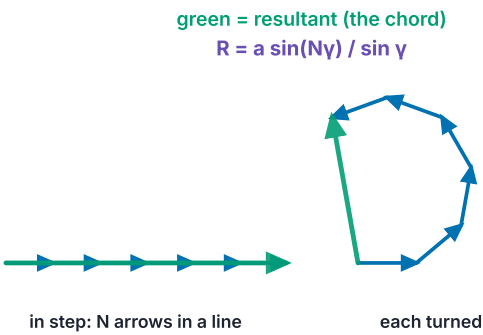
M1 Adding many arrows (phasors)

The picture behind sinβ/β and sinNγ/sinγ.

THE IDEA

Each wave arriving at a point is an arrow: **length** = its amplitude, **angle** = its phase. Waves add like arrows placed head to tail.

If N equal arrows (length a) are each turned a bit more than the last (by the same angle φ), they bend into part of a regular polygon. The total arrow is the **chord** from the first tail to the last head.



N slits as N equal arrows. Each slit's wave is a little further along in phase than the last, so each arrow is turned by the same angle. The arrows form part of a polygon; the resultant is the chord. When the polygon closes, the light cancels.

1. The polygon sits on a circle of radius r. One arrow (a short chord): $a = 2r \sin \dfrac{\phi}{2}$

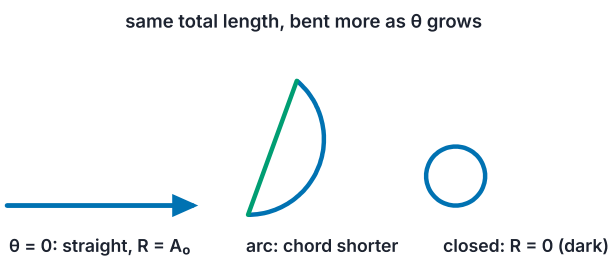
   A chord that subtends angle φ at the centre has length $2r\sin(\phi/2)$.

2. The whole chain subtends $N\phi$, so the total: $R = 2r \sin \dfrac{N\phi}{2}$

3 Divide: $R = a\,\dfrac{\sin(N\phi/2)}{\sin(\phi/2)}$

The famous N-slit factor. When φ = 0 the arrows are in a line and $R = Na$.

Infinitely many tiny arrows



A single slit as infinitely many tiny arrows. They form a smooth arc of fixed length A_0. The resultant is the chord: $A_0 \dfrac{\sin\beta}{\beta}$. The first dark band is where the arc closes into a full circle ($b \sin\theta = \lambda$).

Now let N become huge and each arrow tiny, with the total length A fixed. The polygon becomes a smooth arc of length A. If the arc turns through a total angle 2β, the chord is

$$R = A\,\frac{\sin\beta}{\beta}$$

(the arc's radius is $A/2\beta$, and a chord across angle 2β is $2 \cdot \dfrac{A}{2\beta}\sin\beta$). When the arc closes into a full circle ($2\beta = 2\pi$) the chord is zero: darkness.

> **REMIND ME**
>
> Add arrows first, square at the end. Never add intensities of waves that can interfere.

M2 Small angles, limits and units

> Tools for "what happens at θ = 0" and for grating numericals.

RESULT

For small θ in radians: $\sin\theta \approx \tan\theta \approx \theta$.

$\frac{\sin x}{x} \to 1$ as $x \to 0$. Check: $x = 0.1$: $\sin 0.1/0.1 = 0.9983$.

$\frac{\sin N\gamma}{\sin\gamma} \to \pm N$ as $\gamma \to m\pi$ (both top and bottom → 0; near there $\sin N\gamma \approx N\cdot$small and $\sin\gamma \approx$ small).

1 Å	10^{-8} cm = 10^{-10} m
1 inch	2.54 cm
15000 lines/inch	$d = 2.54/15000 = 1.693 \times 10^{-4}$ cm
6000 lines/cm	$d = 1/6000$ cm

EXAM TRAP

The grating element d is the distance between neighbouring lines = 1 ÷ (lines per unit length). Keep everything in cm or everything in m.

M3 Complex numbers as arrows

> Only needed to read Ghatak's straight-edge and N-slit algebra.

THE IDEA

$e^{i\theta} = \cos\theta + i\sin\theta$ is simply an arrow of length 1 at angle θ (x-part $\cos\theta$, y-part $\sin\theta$). Multiplying by $e^{i\phi}$ turns an arrow by φ. The length of $p + iq$ is $\sqrt{p^2+q^2}$, so its square is p^2+q^2 (that is the intensity).

1. $e^{i0} + e^{i\pi} = 1 + (-1) = 0$
   Two opposite arrows cancel.

2. $\left|\frac{1-i}{2}\right|^2 = \frac{1^2+1^2}{4} = \frac{1}{2}$
   Used in the straight-edge result.

3. Sum of a geometric series: $1 + r + r^2 + \cdots + r^{N-1} = \frac{1-r^N}{1-r}$
   With $r = e^{i\phi}$ this sum of arrows gives the $\sin(N\phi/2)/\sin(\phi/2)$ of M1 again.

M4 Solving tan β = β with a graph

Where are the weak bright bands of a single slit?

THE IDEA

Some equations cannot be solved with algebra. Draw both sides as graphs, $y = \tan\beta$ and $y = \beta$, and read off where they cross. The straight line $y = \beta$ cuts each branch of $\tan\beta$ just before it shoots up at $\frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$

RESULT

Crossings at $\beta = 0,\ 1.43\pi,\ 2.46\pi,\ 3.47\pi,\ \ldots$: slightly **less** than $1.5\pi, 2.5\pi, 3.5\pi$.

REMIND ME · PLUS ONE CALCULUS

A maximum or minimum of $I(\beta)$ is where $dI/d\beta = 0$ (slope zero).

Light bending round corners

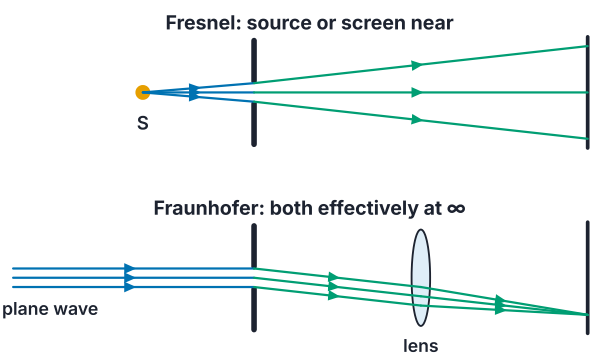
T1 Diffraction; Fresnel and Fraunhofer

What diffraction is, and its two classes (PYQ N25 Q5, N20 Q4).

THE IDEA

When a wave passes an obstacle or an opening, it spreads into the region of the geometrical shadow. This is **diffraction**. The spreading angle is about λ/b, where b is the size of the opening.

Sound ($\lambda \approx 1$ m) bends round buildings easily. Light ($\lambda \approx 5 \times 10^{-5}$ cm) bends only noticeably at openings a fraction of a millimetre wide. That is why we usually think light goes straight.



Two kinds of diffraction. **Fresnel**: the source or the screen is at a finite distance, so the wavefronts are curved. **Fraunhofer**: plane waves in, and the pattern is seen at infinity (in practice, in the focal plane of a lens).

RESULT · TWO CLASSES

Fresnel diffraction: the source, or the screen, or both are at **finite** distances from the aperture. The wavefronts are curved. (Zones, zone plate, straight edge.)

Fraunhofer diffraction: source and screen are effectively at **infinite** distance: plane waves fall on the aperture and the pattern is seen in the focal plane of a lens. (Single slit, double slit, grating.)

RESULT · HUYGENS–FRESNEL PRINCIPLE

Each point on a wavefront is a source of secondary wavelets, and the wavelets from different points interfere with one another. The disturbance at any point is the sum (with phases) of all the wavelets reaching it.

INTERFERENCE VS DIFFRACTION

There is no basic physical difference: both are adding waves. By custom, "interference" means a **few** separate sources (two slits), and "diffraction" means a **continuous** spread of sources across one opening. In a double slit you see both: fringes (interference) inside an envelope (diffraction).

CHECK YOURSELF

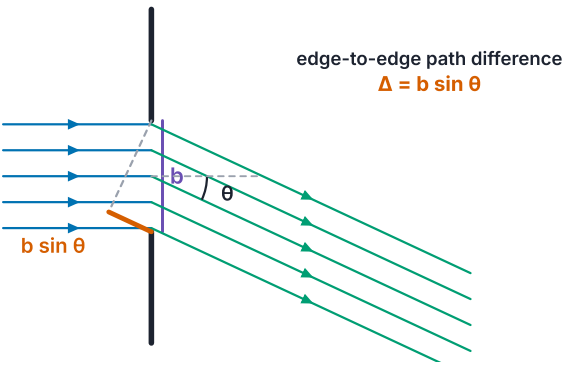
Fill in: In the ____ class of diffraction, the source or screen or both are at finite distances from the aperture.

Fresnel.

In the model paper: **Q1** (PYQ N25 Q5, N20 Q4), **Q2**, **Q3**, **Q6**.

T2 The single slit: intensity formula

> The derivation asked as a 10-mark essay in 2019, 2020, 2022 and 2024.



Single slit of width b, light diffracted at angle θ. The rays from the two edges differ in path by $b\sin\theta$. When this is one whole wavelength, the slit splits into two halves that cancel pair by pair: the first dark band.

A plane wave falls normally on a slit of width b. We look at the light leaving at angle θ, collected by a lens at one point of its focal plane.

1. Divide the slit into n tiny equal strips, spacing Δ, so $b \approx n\Delta$. Each strip sends a wavelet of amplitude a
   Huygens–Fresnel (T1).

2. Neighbouring strips differ in path by $\Delta\sin\theta$, i.e. in phase by $\phi = \dfrac{2\pi}{\lambda}\Delta\sin\theta$
   Path → phase (P1).

3. Add the n arrows, each turned φ from the last: $E_\theta = a\,\dfrac{\sin(n\phi/2)}{\sin(\phi/2)}$
   Phasor chain (M1). Ghatak eq. 18.6.

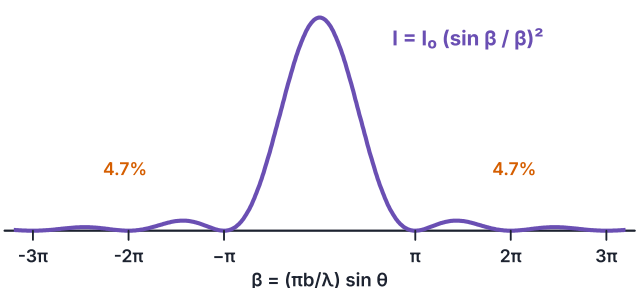
1. Let $n \to \infty$ with $na = A$ fixed. Then $\frac{n\phi}{2} = \frac{\pi n \Delta \sin\theta}{\lambda} \to \frac{\pi b \sin\theta}{\lambda} \equiv \beta$, and $\sin\frac{\phi}{2} \approx \frac{\phi}{2} = \frac{\beta}{n}$
   φ is tiny, so the small-angle rule (M2) applies.

2. $E_\theta = a\,\frac{\sin\beta}{\beta/n} = A\,\frac{\sin\beta}{\beta}$
   Ghatak 18.7. The smooth-arc picture of M1.

3. $I = I_0\,\frac{\sin^2\beta}{\beta^2}$, with $\beta = \frac{\pi b \sin\theta}{\lambda}$
   Square the amplitude. At θ = 0, $\sin\beta/\beta \to 1$, so I_0 is the central intensity (Ghatak 18.10).

RESULT · SINGLE-SLIT FRAUNHOFER PATTERN

$$I(\theta) = I_0\left(\frac{\sin\beta}{\beta}\right)^2, \qquad \beta = \frac{\pi b \sin\theta}{\lambda}$$



Single-slit Fraunhofer pattern. A bright, wide central maximum; dark points at $\beta = \pm\pi, \pm 2\pi, \ldots$ ($b\sin\theta = m\lambda$); weak secondary maxima (the first is only 4.7% of the centre).

SAME RESULT WITH AN INTEGRAL (IF YOUR TEACHER USED IT)

Adding infinitely many strips is an integral: $E \propto \int_{-b/2}^{b/2} e^{ikx\sin\theta}dx = b\,\frac{\sin\beta}{\beta}$. Both routes give the same answer; Ghatak uses the sum.

In the model paper: **Q39** (PYQ N20 Q33 + three more years).

T3 Single slit: dark bands, weak bright bands, width

> "Find the positions of maximum and minimum intensity."

Minima

1. $I = 0$ when $\sin\beta = 0$ but $\beta \neq 0$: $\beta = m\pi$, $m = \pm 1, \pm 2, \ldots$

2. $\dfrac{\pi b \sin\theta}{\lambda} = m\pi \;\Rightarrow\; b\sin\theta = m\lambda$
   Ghatak 18.12. m = 0 is NOT a minimum: it is the central maximum.

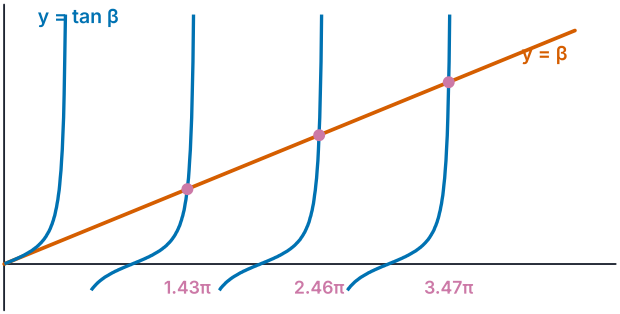
3. Why, in words: at $b\sin\theta = \lambda$, split the slit into two halves. Each strip in the top half has a partner in the bottom half $b/2$ lower, whose path is longer by exactly λ/2. Every pair cancels
   The "arc closes into a circle" picture of M1.

Secondary maxima

1. $\dfrac{dI}{d\beta} = 0$ gives $\sin\beta\,(\beta - \tan\beta) = 0$
   Differentiate $\sin^2\beta/\beta^2$ (M4).

2. $\sin\beta = 0$ gives the minima; $\tan\beta = \beta$ gives the maxima: $\beta = 0, 1.43\pi, 2.46\pi, 3.47\pi, \ldots$

3. Their intensities: I_0, $0.047I_0$, $0.016I_0$, $0.008I_0$
   The first side band is only about 1/21 of the centre.



The secondary maxima of a single slit are where the line $y = \beta$ meets $y = \tan\beta$ (pink dots): at $\beta = 1.43\pi, 2.46\pi, \ldots$, a little before $1.5\pi, 2.5\pi, \ldots$.

Width of the central maximum

1. First zeros at $\sin\theta = \pm\lambda/b$, so the central band spans $-\lambda/b \le \theta \le \lambda/b$
   Angular half-width λ/b.

2. On a screen in the focal plane of a lens of focal length f: total width $= \dfrac{2f\lambda}{b}$

RESULT

Minima	$b\sin\theta = m\lambda,\ m = \pm1, \pm2, \ldots$
Secondary maxima	$\tan\beta = \beta;\ \beta \approx \pm1.43\pi, \pm2.46\pi$
Central width	$2\lambda/b$ (angle), $2f\lambda/b$ (on screen)

Narrower slit or longer wavelength → **wider** pattern.

1. Example: $b = 0.2$ mm, λ = 5000 Å, $f = 20$ cm: $\lambda/b = \dfrac{5\times10^{-5}}{0.02} = 2.5\times10^{-3}$ rad

2. First minimum $20\times2.5\times10^{-3} = 0.05$ cm from the centre; central band 0.1 cm wide; second minimum at 0.1 cm

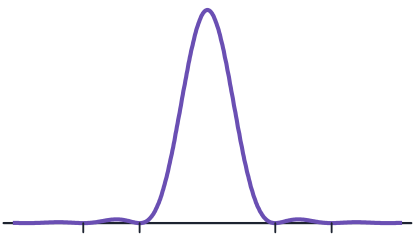
EXAM TRAP

The secondary maxima are **not** exactly halfway between minima (not at 1.5π); they are at 1.43π, 2.46π, ...

In the model paper: **Q4** , **Q5** , **Q38** , **Q39** .

T4 A round hole, and the limit of resolution

> Asked in 2019, 2020, 2022 though not in the listed sections (Ghatak §18.3, 18.5).





Fraunhofer diffraction by a circular hole of diameter D: a bright central disc (the Airy disc, about 84% of the light) surrounded by faint rings. The first dark ring is at $\sin\theta = 1.22\lambda/D$.

RESULT · FRAUNHOFER DIFFRACTION AT A CIRCULAR APERTURE

Pattern: concentric rings round a bright central disc (Airy pattern). $I = I_0\left[\frac{2J_1(v)}{v}\right]^2$, $v = \frac{2\pi a}{\lambda}\sin\theta$ (a = radius).

Dark rings at $v = 3.832, 7.016, \ldots$, i.e. first dark ring at $\sin\theta = \frac{0.61\lambda}{a} = \frac{1.22\lambda}{D}$. About 84% of the light is in the central disc. Smaller hole → bigger disc.

RESULT · RAYLEIGH CRITERION AND RESOLUTION

Two point sources are just resolved when the central maximum of one falls on the first minimum of the other. For a lens or eye of diameter D, the smallest resolvable angle is

$$\Delta\theta_{min} = \frac{1.22\lambda}{D}$$

1. Eye (PYQ N19 Q28): D = 2 mm, λ = 6×10^{-7} m:
$$\Delta\theta = \frac{1.22 \times 6 \times 10^{-7}}{2 \times 10^{-3}} = 3.66 \times 10^{-4} \text{ rad}$$

2. At 30 m: separation $= 30 \times 3.66 \times 10^{-4} = 0.011$ m ≈ 1.1 cm

MICROSCOPE AND OIL IMMERSION (PYQ N19 Q19)

A microscope can separate two points a distance $d_{min} = \frac{0.61\lambda}{n \sin\alpha}$ apart, where α is the half-angle of the cone of light entering the objective and n is the refractive index between object and lens. $n \sin\alpha$ is the **numerical aperture**. Filling that space with oil ($n \approx 1.5$) instead of air raises $n \sin\alpha$, so d_{min} falls: finer detail is seen. (Shorter wavelength helps too.)

LOUDSPEAKER (PYQ N19 Q12)

The sound spreads over an angle of about $1.22\lambda/D$. To make the sound more directional (a narrow beam), use a loudspeaker of **larger diameter** compared with the wavelength.

EXAM TRAP

For a round hole use $1.22\lambda/D$, not λ/D (that is for a slit).

In the model paper: **Q7** (PYQ N19 Q7), **Q8** (PYQ N19 Q12), **Q29** (PYQ N19 Q19), **Q30** (PYQ N19 Q28), **Q31** (PYQ N20 Q21, N22 Q22).

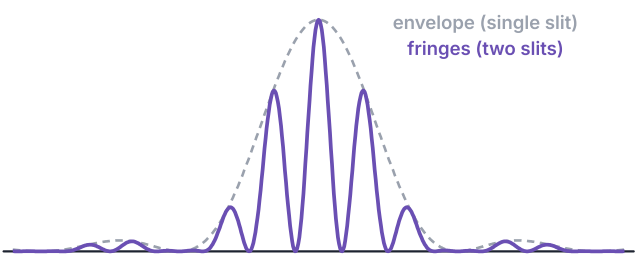
T5 Two slits: fringes inside an envelope

> Young's fringes and single-slit diffraction at the same time.

1. Each slit (width b) gives amplitude $A\dfrac{\sin\beta}{\beta}$ at angle θ
2. The second slit is d further on (centre to centre), so its wave is behind by phase 2γ, with $\gamma = \dfrac{\pi d \sin\theta}{\lambda}$
3. Two equal arrows at angle 2γ: total $2A\dfrac{\sin\beta}{\beta}\cos\gamma$

   Isosceles triangle: resultant $= 2\times$side$\times\cos($half the angle$)$.
4. $I = 4I_0\,\dfrac{\sin^2\beta}{\beta^2}\cos^2\gamma$

   Ghatak 18.45: single-slit envelope × Young fringes.



here d = 3b: the 3rd, 6th ... bright fringes are missing

Two slits, each of width b, centres d apart: $I = 4I_0\left(\dfrac{\sin\beta}{\beta}\right)^2\cos^2\gamma$.

Fine interference fringes ($\cos^2\gamma$) sit inside the single-slit envelope. Where a fringe falls on an envelope zero, it is a **missing order**.

RESULT · TWO SLITS

Bright fringes	$d\sin\theta = m\lambda$
Dark fringes	$d\sin\theta = (m+\frac{1}{2})\lambda$
Envelope zeros	$b\sin\theta = p\lambda$

RESULT · MISSING ORDERS

A bright fringe is **missing** when it falls exactly on an envelope zero, i.e. when $d\sin\theta = m\lambda$ and $b\sin\theta = p\lambda$ at the same θ:

$$\frac{d}{b} = \frac{m}{p}$$

If $d = 3b$: orders 3, 6, 9, … are missing, and the central envelope holds $2(d/b) - 1 = 5$ fringes.

EXAM TRAP

The fringe spacing depends on the slit **separation** d; the slit **width** b only sets the envelope.

In the model paper: **Q9**, **Q26**.

T6 N slits: the intensity distribution

> PYQ N25 Q21 (10 marks). The same arrow chain, now with N slits.

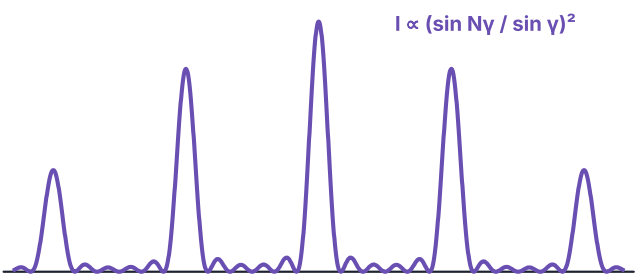
1. Each of the N slits gives an arrow of length $A\dfrac{\sin\beta}{\beta}$; neighbours differ in phase by 2γ, $\gamma = \dfrac{\pi d \sin\theta}{\lambda}$

2. Chain of N arrows, each turned 2γ: total $A\dfrac{\sin\beta}{\beta} \cdot \dfrac{\sin N\gamma}{\sin\gamma}$

   M1 with $\phi = 2\gamma$. Ghatak 18.49.

3. $I = I_0 \dfrac{\sin^2\beta}{\beta^2} \cdot \dfrac{\sin^2 N\gamma}{\sin^2\gamma}$

   Ghatak 18.50. N = 1 gives the single slit; N = 2 gives T5 (since $\sin 2\gamma / \sin\gamma = 2\cos\gamma$).



N = 6: sharp principal maxima, N − 2 = 4 tiny ones between

N slits. The principal maxima stay at the same places ($d\sin\theta = m\lambda$) but become taller ($\propto N^2$) and much narrower as N grows. Between two principal maxima there are $N-1$ minima and $N-2$ weak secondary maxima.

RESULT · FEATURES OF THE N-SLIT PATTERN

Principal maxima	$\gamma = m\pi$, i.e. $d\sin\theta = m\lambda$. There $\frac{\sin N\gamma}{\sin\gamma} \to N$, so $I = N^2 I_0 \frac{\sin^2\beta}{\beta^2}$: all slits in step
Minima	$N\gamma = p\pi$, $p \neq 0, N, 2N, \ldots$: $d\sin\theta = \frac{\lambda}{N}, \frac{2\lambda}{N}, \ldots, \frac{(N-1)\lambda}{N}, \ldots$
Between two principal maxima	$N-1$ minima and $N-2$ weak secondary maxima
Missing orders	as for two slits, when $b\sin\theta = p\lambda$ coincides

WHY PRINCIPAL MAXIMA GET SHARP

Height grows as N^2 but the first zero beside each peak moves in to λ/N. So with many slits the bright lines become very tall and very thin (T7).

EXAM TRAP

"Principal maximum becomes sharper as N **increases**": true (N22 Q10). "... as N decreases": false (N19 Q6).

In the model paper: **Q10** (PYQ N22 Q10, N19 Q6), **Q11** , **Q40** (PYQ N25 Q21).

T7 Width of a principal maximum

> PYQ N24 Q17: "Derive the equation for width of principal maxima of an N slit Fraunhofer diffraction."

1. The m-th principal maximum is at $d \sin\theta_m = m\lambda$, i.e. $N\gamma = mN\pi$
   Ghatak 18.60.

2. The nearest zeros on either side are at $N\gamma = (mN \pm 1)\pi$, i.e. $d\sin(\theta_m \pm \Delta\theta_m) = m\lambda \pm \dfrac{\lambda}{N}$
   Ghatak 18.61.

3. $\sin(\theta_m \pm \Delta\theta_m) \approx \sin\theta_m \pm \Delta\theta_m \cos\theta_m$
   Δθ is tiny: $\sin(A + x) \approx \sin A + x\cos A$ (18.62).

4. Subtract step 1: $d\,\Delta\theta_m \cos\theta_m = \dfrac{\lambda}{N}$

5. $\Delta\theta_m = \dfrac{\lambda}{Nd\cos\theta_m}$
   Ghatak 18.63: angular half-width. Full width $2\Delta\theta_m$.

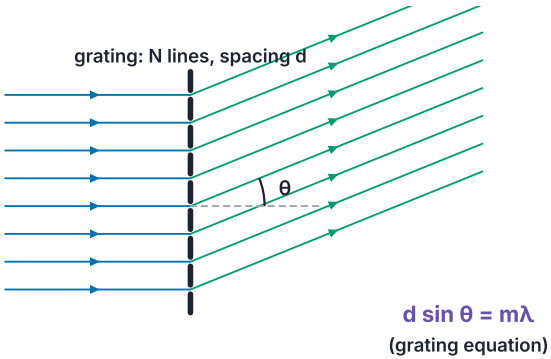
RESULT

$\Delta\theta_m = \dfrac{\lambda}{Nd\cos\theta_m}$. Nd is the total width of the grating, so a wider grating (more lines) gives sharper lines.

In the model paper: **Q25** (PYQ N24 Q17).

T8 The diffraction grating

> Many equally spaced slits: a tool that spreads light into its colours.



A diffraction grating: many equally spaced slits. Bright lines appear only in directions where light from neighbouring slits differs in path by a whole number of wavelengths.

RESULT · GRATING (PYQ N20 Q15)

A diffraction grating is a plate with a very large number of equally spaced parallel slits (lines ruled on glass, thousands per cm). The distance d from one line to the next is the **grating element**. Light of wavelength λ gives bright principal maxima where

$$d \sin\theta = m\lambda \qquad (m = 0, 1, 2, \ldots)$$

$m = 0$ is at θ = 0 for every colour (white). Each higher order spreads white light into a spectrum, violet nearest the centre.

1. Highest order: $\sin\theta \leq 1$, so $m \leq d/\lambda$. Example: 5000 lines/cm, λ = 5000 Å: $d = 2 \times 10^{-4}$ cm, $d/\lambda = 4$
   m = 4 would need θ = 90° (grazing), which cannot be observed, so the highest visible order is 3.

Dispersion

1. Differentiate $d \sin\theta = m\lambda$ at fixed m: $d\cos\theta\,\Delta\theta = m\,\Delta\lambda$
2. $$\frac{\Delta\theta}{\Delta\lambda} = \frac{m}{d\cos\theta}$$
   Ghatak 18.65: angular dispersion. Bigger for higher order, and for **smaller** d (finer grating).

1. PYQ N20 Q27: 15000 lines/inch, $d = 1.693 \times 10^{-4}$ cm; sodium λ = 5890, 5896 Å; f = 10 cm. First order: $\sin\theta = \dfrac{5.89 \times 10^{-5}}{1.693 \times 10^{-4}} = 0.348$, $\cos\theta = 0.938$
2. $$\Delta\theta = \frac{1 \times 6 \times 10^{-8}}{1.693 \times 10^{-4} \times 0.938} = 3.78 \times 10^{-4} \text{ rad}$$
3. On the screen: $f\Delta\theta = 10 \times 3.78 \times 10^{-4} = 3.8 \times 10^{-3}$ cm ≈ 0.038 mm

OVERLAPPING ORDERS (PYQ N22 Q15)

Order m of λ_2 lands on order m+1 of λ_1 when $m\lambda_2 = (m+1)\lambda_1$. For visible light (400–700 nm) the 1st and 2nd orders never overlap (2 × 400 = 800 > 700), but the 2nd and 3rd do (3 × 400 = 1200 < 2 × 700 = 1400).

GRATING VS PRISM (PYQ N20 Q7)

Grating spectra are purer and sharper, and nearly linear (Δθ ∝ Δλ for small θ). A prism's spread depends on how n changes with λ, which is uneven. So "grating spectra are purer than prism spectra" is **true**.

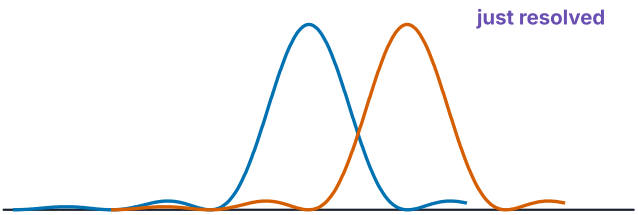
EXAM TRAP

"Larger the grating element, larger the dispersion": **false** (N19 Q9). Dispersion ∝ 1/d.

In the model paper: **Q12** (PYQ N20 Q15), **Q13** (PYQ N19 Q9), **Q14** (PYQ N22 Q15), **Q15** (PYQ N20 Q7), **Q18**, **Q28** (PYQ N20 Q27).

T9 Resolving power of a grating

> Can the grating separate two very close wavelengths? Asked in 2022, 2024, 2025.



max of one falls on the first minimum of the other

Rayleigh criterion: two wavelengths are "just resolved" when the principal maximum of one falls on the first minimum of the other. For a grating this gives the resolving power $\lambda/\Delta\lambda = mN$.

1. m-th order of λ: $d\sin\theta = m\lambda$. Of $\lambda+\Delta\lambda$: $d\sin\theta' = m(\lambda+\Delta\lambda)$
2. Rayleigh: just resolved when the maximum of $\lambda+\Delta\lambda$ falls on the first minimum next to the maximum of λ, which is at $d\sin\theta = m\lambda + \dfrac{\lambda}{N}$

   The first zero beside a principal maximum (T6, T7).
3. $m(\lambda+\Delta\lambda) = m\lambda + \dfrac{\lambda}{N} \;\Rightarrow\; m\Delta\lambda = \dfrac{\lambda}{N}$
4. $R = \dfrac{\lambda}{\Delta\lambda} = mN$

   Ghatak 18.69. N = total number of lines illuminated.

RESULT

$R = \dfrac{\lambda}{\Delta\lambda} = mN$. It depends on the order and the total number of lines, not on d.

1. PYQ N22 Q29: 0.15 m ruled with 6×10^5 lines/m: $N = 0.15 \times 6 \times 10^5 = 90000$; first order $R = 90000$

2. PYQ N25 Q15: 1 inch, 15000 lines, first order: $R = 15000$; $\Delta\lambda = \frac{500}{15000} = 0.033$ nm. Next resolvable wavelength ≈ 500.033 nm

COMPARE A PRISM

A prism's resolving power is $t\,\frac{dn}{d\lambda}$ (t = base length). A 2.5 cm crown-glass prism gives only about 1000.

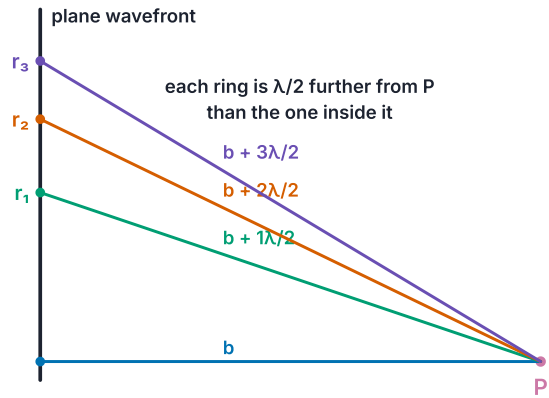
EXAM TRAP

Resolving power is $\lambda/\Delta\lambda$ (a big number), not $\Delta\lambda/\lambda$.

In the model paper: **Q16** (PYQ N24 Q8, N22 Q5), **Q17**, **Q27** (PYQ N22 Q29, N25 Q15).

T10 Fresnel's half-period zones

> Cut the wavefront into rings that arrive alternately in and out of step. PYQ N19 Q20, N24 Q19, N25 Q6.



Fresnel half-period zones. Divide the wavefront into rings whose distances to P increase by λ/2 each time. Light from neighbouring zones arrives half a wave apart (out of step), so their effects nearly cancel. $r_m \approx \sqrt{m\lambda b}$.

1. Plane wavefront WW′; point P at distance d from it (O is the foot of the perpendicular). Draw spheres about P of radii $d + \frac{\lambda}{2}, d + \frac{2\lambda}{2}, d + \frac{3\lambda}{2}, \ldots$. They cut the wavefront in circles: the zones

2. Radius of the n-th circle: $r_n^2 = \left(d + \frac{n\lambda}{2}\right)^2 - d^2 = n\lambda d + \frac{n^2\lambda^2}{4} \approx n\lambda d$
   Pythagoras; λ ≪ d so drop the λ^2 term. $r_n = \sqrt{n\lambda d}$ (Ghatak 20.1).

3. Area of the n-th zone: $\pi r_n^2 - \pi r_{n-1}^2 = \pi\lambda d$
   Independent of n: **all zones have (almost) equal area** (20.2).

EXACT AREA DIFFERENCE (PYQ N25 Q6)

Keeping the λ² term: area of zone n $= \pi\lambda d + \frac{\pi\lambda^2(2n-1)}{4}$. So (area of zone n) − (area of zone n−1) $= \frac{\pi\lambda^2}{2}$, which is negligibly small: the areas are practically equal.

1. Light from neighbouring zones arrives with path difference λ/2, i.e. phase π: their effects at P have opposite signs. Amplitude $u = u_1 - u_2 + u_3 - u_4 + \cdots$

   "Half-period": half a period apart in time.

2. Equal areas, but each zone is a little further from P and more tilted, so $u_1 > u_2 > u_3 > \cdots$, decreasing slowly

3. Group as $\frac{u_1}{2} + \left(\frac{u_1}{2} - u_2 + \frac{u_3}{2}\right) + \left(\frac{u_3}{2} - u_4 + \frac{u_5}{2}\right) + \cdots$. Each bracket ≈ 0, because each u_n is close to the average of its neighbours

4. $u \approx \frac{u_1}{2}$

   The whole wavefront gives half the amplitude of the first zone alone (Ghatak 20.12). Intensity is ¼ of the first zone's.

EXAM TRAP

Zone **radii** grow as $\sqrt{n}$ (not equally spaced); zone **areas** are equal.

In the model paper: **Q19** (PYQ N25 Q6), **Q20**, **Q32** (PYQ N19 Q20, N24 Q19).

T11 Straight-line travel, holes and discs

PYQ N22 Q34: rectilinear propagation from half-period zones.

RECTILINEAR PROPAGATION

The light at P is $u_1/2$: it effectively comes only from the first zone (and its neighbours), a tiny disc of radius $\sqrt{\lambda d} \approx 0.5$ mm for d = 50 cm, λ = 5000 Å. The outer zones cancel each other in pairs.

So an obstacle much larger than this tiny region throws a sharp shadow, and light seems to travel in straight lines. Only openings or obstacles comparable to $\sqrt{\lambda d}$ show noticeable diffraction.

RESULT · SMALL CIRCULAR HOLE (RADIUS A), ON THE AXIS

The hole lets through $p = \dfrac{a^2}{\lambda d}$ zones. Odd p: bright (about u_1, i.e. 4 times the unobstructed intensity). Even p: nearly dark. Brightest at $d = a^2/\lambda$ (one zone); darkest at $d = a^2/2\lambda$ (two zones).

1. a = 0.5 mm, λ = 5×10^{-5} cm: brightest at $d = \dfrac{0.0025}{5 \times 10^{-5}} = 50$ cm; darkest at 25 cm

OPAQUE DISC: POISSON'S BRIGHT SPOT

A small disc blocks the first p zones. The rest give $u_{p+1} - u_{p+2} + \cdots \approx u_{p+1}/2$, which is never zero. So the **centre of the shadow is bright**. Poisson (1818) used this to mock Fresnel's theory; Arago then saw the spot, a famous win for the wave theory.

In the model paper: Q21, Q33, Q41 (PYQ N22 Q34).

T12 The zone plate

> Block every other zone and you get a lens made of rings. PYQ N22 Q21, N19 Q29, N20 Q30.



every other zone blocked

only zones that add in step are left open → bright focus

$$f = r_1^2 / \lambda$$

also weaker foci at f/3, f/5, ...

A zone plate: rings with radii $\propto \sqrt{m}$, alternate rings blackened. All open zones send light to P in step, so P is very bright. It acts like a lens with focal length r_1^2/λ.

RESULT · PRINCIPLE AND CONSTRUCTION

Draw circles of radii proportional to $\sqrt{1}, \sqrt{2}, \sqrt{3}, \ldots$ on white paper and blacken alternate rings; photograph it small on glass. For a point P at distance $f = r_1^2/\lambda$, the rings are exactly its half-period zones. The open (odd) zones all send light in step: $u = u_1 + u_3 + u_5 + \cdots$, so P is very bright.

1. Lens formula: source S at a, image at b. Edge of zone 1 at height r_1: path S→edge→P is λ/2 longer than the straight path: $\frac{r_1^2}{2}\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{\lambda}{2}$

   $\sqrt{a^2 + r^2} \approx a + r^2/2a$ (binomial).

2. $\frac{1}{a} + \frac{1}{b} = \frac{\lambda}{r_1^2} = \frac{1}{f}$, with $f = \frac{r_1^2}{\lambda}$

   Same form as a lens (Ghatak 20.20).

MORE FOCI

At $f/3$ each open ring holds three zones, so it still adds up; also at $f/5, f/7, \ldots$ (weaker). Between them (at $f/2, f/4$) there are dark points.

1. PYQ N19 Q29: $r_n = 0.001\sqrt{n}$ m, λ = 4.5×10^{-7} m: $f = \dfrac{(10^{-3})^2}{4.5 \times 10^{-7}} = 2.22$ m. Other foci: 0.741, 0.444, 0.317 m

2. PYQ N20 Q30: $f = 60$ cm, λ = 6000 Å: $r_1 = \sqrt{f\lambda} = \sqrt{60 \times 6 \times 10^{-5}} = 0.06$ cm = 0.6 mm

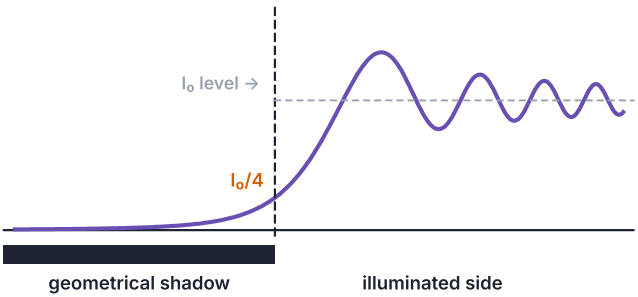
RESULT · ZONE PLATE VS CONVEX LENS

Working	diffraction (zone plate) vs refraction (lens)
Foci	many ($f, f/3, f/5, \ldots$) vs one
Colour	$f \propto 1/\lambda$: red focuses **nearer** than violet; for a glass lens violet focuses nearer
Brightness	half the area is blocked, so less bright

In the model paper: **Q22** , **Q34** (PYQ N22 Q21), **Q35** (PYQ N19 Q29, N20 Q30), **Q36** .

T13 The straight edge and the Cornu spiral

Why a shadow is never perfectly sharp (Ghatak §20.6).



Diffraction at a straight edge. In the shadow the light fades smoothly to zero; at the edge of the shadow it is exactly $I_0/4$; on the bright side it wobbles above and below I_0 (first peak ≈ 1.37 I_0) before settling.

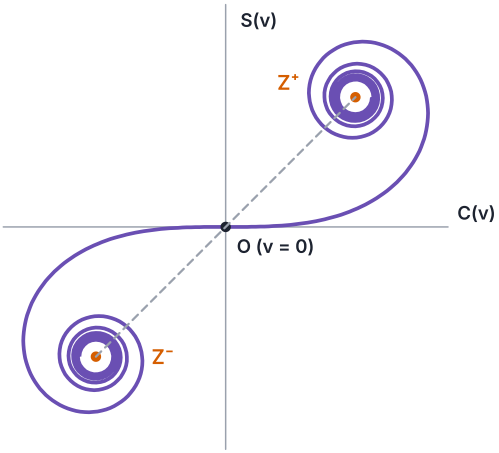
THE IDEA

A straight edge blocks half the wavefront. For a point P on the screen, we add up the light from the open half, strip by strip. Strips far from P's line contribute arrows that turn faster and faster, so the arrows curl into a double spiral: the **Cornu spiral**.

RESULT · FRESNEL INTEGRALS

$$C(v)=\int_0^v \cos\frac{\pi u^2}{2}\,du, \qquad S(v)=\int_0^v \sin\frac{\pi u^2}{2}\,du$$

Both are 0 at $v=0$ and tend to $\pm\frac{1}{2}$ as $v\to\pm\infty$. The curve $(C(v),S(v))$ is the Cornu spiral, winding into the points $Z^{\pm}=(\pm\frac{1}{2},\pm\frac{1}{2})$.



The Cornu spiral: the points $(C(v), S(v))$ of the Fresnel integrals. The amplitude at a point of the screen is the length of the chord between the right two points of the spiral. Intensity ∝ (chord)².

1. For a screen point P at height y (shadow edge at $y = 0$), set $v_0 = -\sqrt{\dfrac{2}{\lambda d}}\, y$ (d = distance edge → screen)

2. Amplitude ∝ the chord from the spiral point v_0 to Z^+ : $I = \dfrac{I_0}{2}\left\{\left[\tfrac{1}{2} - C(v_0)\right]^2 + \left[\tfrac{1}{2} - S(v_0)\right]^2\right\}$
   Ghatak 20.58. I_0 = intensity without the edge (chord $Z^- Z^+$).

3. At the shadow edge, $v_0 = 0$: $I = \dfrac{I_0}{2}\left(\dfrac{1}{4} + \dfrac{1}{4}\right) = \dfrac{I_0}{4}$
   The chord is half the full one, so a quarter of the intensity.

RESULT · THE PATTERN

Deep shadow	falls smoothly to 0 (no fringes)
Shadow edge	$I_0/4$
Bright side	fringes: maxima at $\lvert v_0\rvert$ = 1.22, 2.34, 3.08 ($1.37I_0$, $1.20I_0$, $1.15I_0$), minima at 1.87, 2.74, …; they crowd together and settle to I_0
Positions	$y = \lvert v_0\rvert\sqrt{\lambda d/2}$

1 λ = 6×10^{-5} cm, d = 120 cm: $\sqrt{\lambda d/2} = 0.06$ cm. First two maxima at $1.22 \times 0.06 = 0.073$ cm and $2.34 \times 0.06 = 0.140$ cm from the edge of the shadow

In the model paper: **Q23** , **Q24** , **Q37** , **Q42** .

One-page summary

Fresnel	finite distances, curved wavefronts
Fraunhofer	plane waves, pattern at ∞ (lens)
Single slit	$I = I_0 \frac{\sin^2 \beta}{\beta^2}$, $\beta = \frac{\pi b \sin\theta}{\lambda}$
Minima	$b \sin\theta = m\lambda$ ($m \neq 0$)
Sec. maxima	$\tan\beta = \beta$: 1.43π, 2.46π; 4.7%, 1.6%
Round hole	first dark ring $1.22\lambda/D$
Two slits	$4I_0 \frac{\sin^2 \beta}{\beta^2} \cos^2\gamma$, $\gamma = \frac{\pi d \sin\theta}{\lambda}$
Missing order	$d/b = m/p$
N slits	$I_0 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma}$; peaks N^2
Peak width	$\Delta\theta = \frac{\lambda}{N d \cos\theta}$
Grating	$d \sin\theta = m\lambda$; $m \leq d/\lambda$
Dispersion	$\frac{\Delta\theta}{\Delta\lambda} = \frac{m}{d\cos\theta}$
Resolving power	$\frac{\lambda}{\Delta\lambda} = mN$
Zones	$r_n = \sqrt{n\lambda d}$, area $\pi\lambda d$; total $u_1/2$
Zone plate	$f = r_1^2/\lambda$; also $f/3, f/5$
Straight edge	$I_0/4$ at edge; first max $1.37 I_0$

Exam question → topics to know

PYQ	QUESTION (SHORT)	KNOW
N25 Q5, N20 Q4	Fresnel vs Fraunhofer	T1
N19 Q33, N20 Q33, N22 Q33, N24 Q21	Single slit: pattern, maxima, minima	T2 T3 M1 M4
N19 Q7, N20 Q21, N22 Q22	Circular aperture	T4
N19 Q12, Q19, Q28	Loudspeaker; microscope; eye	T4
N25 Q21	N-slit intensity	T6 M1
N19 Q6, N22 Q10	Sharpness vs N (T/F)	T6 T7
N24 Q17	Width of principal maxima	T7
N20 Q15, Q7; N19 Q9; N22 Q15	Grating; purity; dispersion; overlap	T8
N20 Q27	Sodium D-line separation	T8 M2
N22 Q5, N24 Q8, N22 Q29, N25 Q15	Resolving power of grating	T9
N19 Q20, N24 Q19, N25 Q6	Half-period zones, areas	T10
N22 Q34	Rectilinear propagation	T10 T11
N22 Q21, N19 Q29, N20 Q30	Zone plate	T12

No past paper has asked the straight edge (T13) or two slits (T5) yet: they are in the syllabus, so the model paper covers them.